DSA 107 · DATA SCIENCE AND ANALYTICS LABORATORY

Lab 2 Report

Summarising One Variable, Then Two

Class, survival and age in the Titanic passenger list

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1. Objective

This second lab moved from looking at a table to summarising what is inside it. The aim was to practise the right summary for each kind of variable: counts and proportions for one categorical variable; a crosstab and, from it, a conditional table for two categorical variables together; and centre plus spread for a quantitative variable - with each summary drawn as a labelled figure, three in total. My own test for the session: to be able to explain why the same six numbers become different tables depending on what you divide by.

2. Data Description

The dataset is a passenger list of the Titanic, read from the course's teaching copy in the seaborn-data repository directly from its URL with pd.read_csv, so anyone who runs the notebook gets the same 891 rows. Each row is one passenger, described by 15 columns: the outcome flag survived, the ticket class pclass, sex, age in years, the fare paid, and family and boarding details. pclass is stored as the numbers 1-3 but is categorical: second class is not twice first class, the number is a label.

Nothing was cleaned and all rows were kept. The missing values are not spread evenly: deck is missing in 688 of the 891 rows (77%), age in 177, and embarked and embark_town in 2 each. A column missing three-quarters of its values is not usable as a category, so deck plays no part in this analysis; the readings below work with each column's available values. Table 1 collects the details.

Property	Value
Source	seaborn-data teaching copy of the Titanic passenger list, read from github.com/mwaskom/seaborn-data (titanic.csv)
Size	891 rows × 15 columns (one passenger per row)
Outcome variable	survived - a 0/1 flag, compared across classes; no modelling in this lab
Key features	categorical: pclass, sex, embarked; numeric: age (years), fare, sibsp, parch (counts); text versions: class, who, alive
Missing values	deck: 688 rows (77%); age: 177; embarked and embark_town: 2 each - none dropped
After preprocessing	no cleaning applied; all 891 rows retained

Table 1. Dataset summary and what was (not) done to it.

3. Methodology

One Colab notebook, run top to bottom after a restart, with pandas 2.2.3 plus matplotlib and seaborn (whitegrid). The steps, in order, with the reason each was chosen:

1. **Load and look.** read_csv(url) and head() check the read; .shape (891, 15) and .info() give the types and non-missing counts; isna().sum().sort_values(ascending=False) puts the worst column first instead of hunting for it.
2. **One categorical variable.** counts_class = titanic["pclass"].value_counts().sort_index() - sorted by class number, because bars ordered 2-1-3 by count would misread the chart; the normalized counts give proportions, checked to sum to 1.
3. **Two categorical variables.** pd.crosstab(pclass, survived), then row totals with sum(axis="columns") checked against len(titanic); the denominator step: counts.div(row_totals, axis="index"), checked to give 1.0 on every row.
4. **One quantitative variable.** age as a histogram (10 bins) - value_counts would give 88 rows with no visible shape; mean, median and std for age, and mean, median, max for fare, because the median-mean gap exposes a skew that one average would hide.
5. **Save.** the finished notebook was pushed to my dsa107-labs repository on GitHub.

4. Results

Figure 1 gives the class distribution: 491 of the 891 passengers travelled third class (55.1%), against 216 in first (24.2%) and 184 in second (20.7%). The pair pclass × survived counts 372 third-class deaths - the largest single number in the table, but a raw count cannot compare groups of different sizes.

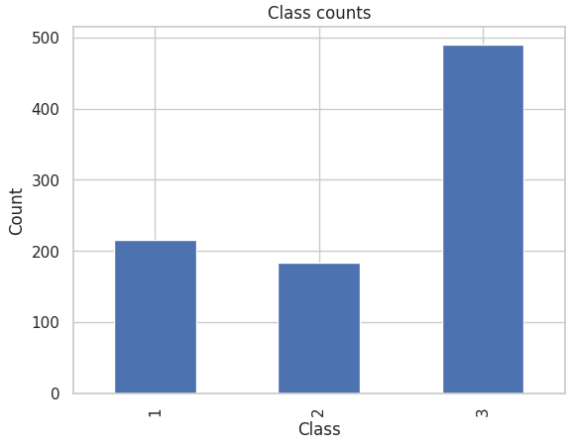


Figure 1. Passengers per ticket class (n = 891).

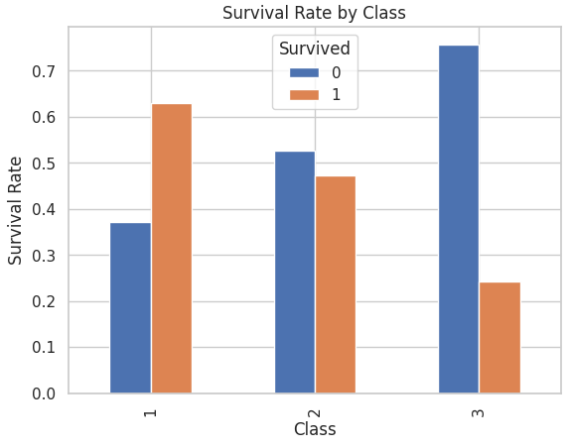


Figure 2. Survival and death rates within each class.

Dividing every cell by its row total makes the comparison fair. Given first class, 63.0% survived and 37.0% died; given third class only 24.2% survived (119 of 491) against 75.8% who died; second class sits between, at 47.3%. A first-class passenger was about 2.6 times as likely to survive as a third-class one.

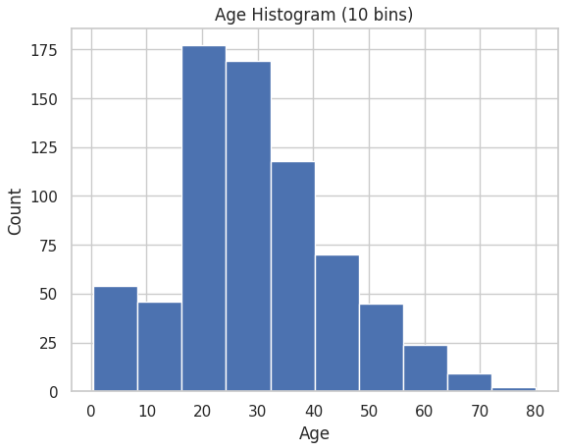


Figure 3. Age distribution, 10 bins (177 rows without age).

Figure 3 shows where the age data piles up: the biggest bars fall between the mid-20s and the early 30s (mean 29.7, median 28.0, standard deviation 14.5). Fare has the opposite shape - median 14.45 but mean 32.20, with a maximum of 512.33 - a few very expensive tickets pull the mean up, so the median is the honest typical fare.

5. Discussion

Class and survival are clearly linked here, and the link survives the fair-comparison test: the gap between 63.0% and 24.2% is not an artefact of third class being the largest group - that is exactly what the conditional table was for, because every row adds up to 1 whatever its size. The pattern does not say why class mattered: the data records who lived, not where anyone sat or who reached a boat, so this is an association, not an explanation. Two further limits: any age-based reading silently drops the 177 passengers with no recorded age, and this analysis never separated men from women - a crosstab of survival within sex is the first thing I would check next.

6. Reflection

What was difficult was the denominator idea, and my first answer shows it: I started from "372 is the biggest number, so third class was the most dangerous", then corrected myself in the same answer - a count without its row total compares nothing, because 372 out of 491 is a very different sentence from 372 out of 891. Dividing the crosstab row by row and watching every row come back to 1.0 is what made conditional distributions click. A smaller surprise was .info() versus .describe(): the first prints instead of returning, the second hands back a table you can keep. What I understand now that I did not before: the same six counts really are three different tables, and the question chooses which one to build - counts show who is in the data, rates compare the groups fairly.

7. References

[1] Seaborn-data teaching copy of the Titanic passenger list (titanic.csv), github.com/mwaskom/seaborn-data - the file read by the notebook; original distribution: Titanic - Machine Learning from Disaster, Kaggle, kaggle.com/c/titanic.

[2] Waskom, M. (2021). seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021.